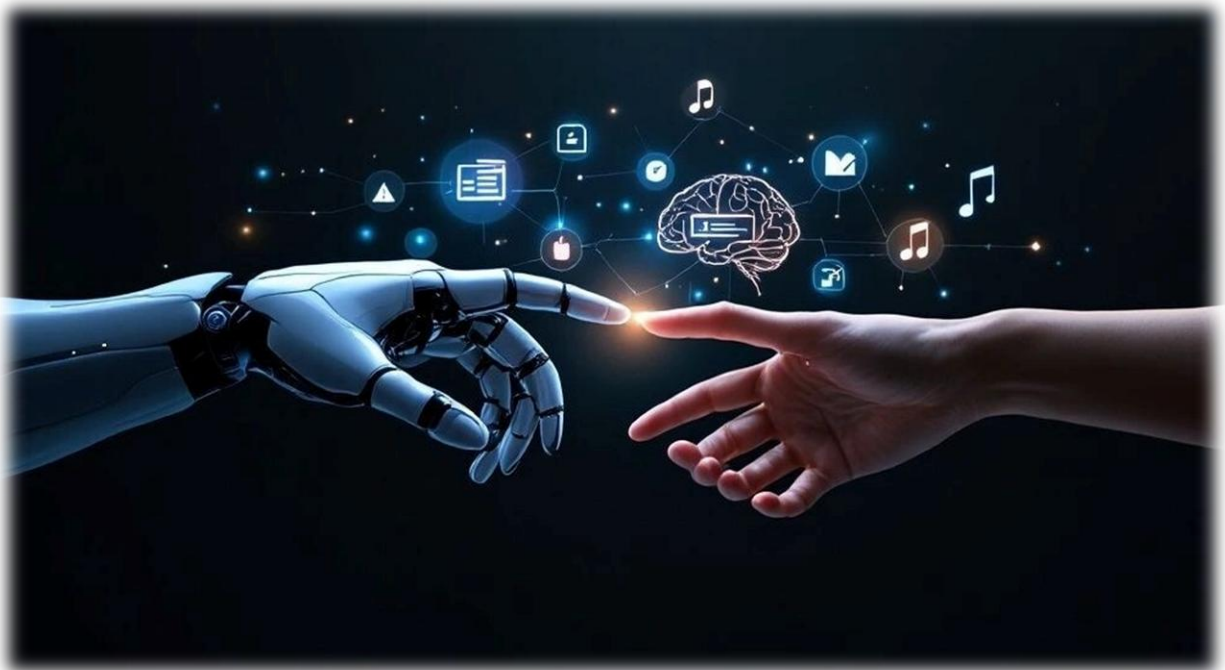




AIR UNIVERSITY, ISLAMABAD



GENERATIVE AI FOR SOFTWARE ENGINEERING

SUBMITTED TO: DR HUMAIRA WAQAS

SUBMITTED BY: ALIZA ZAMAN | 232016-BSSE-6B | 2026

LAB 06

Task 1:

Add a Wikipedia Summary Tool

Fourth Tool Added:

Agentic RAG Chatbot

Lab 06 — 4 tools: Knowledge Base | Web Search | Calculator | Wikipedia

Health CheckIngest PDFList ToolsClear Chat

Welcome! Checking server connection...

Status: ok | Vectors: 27 | Web Search: true | LLM: llama-3.1-8b-instant

PDF ingested successfully!
Chunks: 24
File:

Available tools:

- search_knowledge_base: Search PDF knowledge base in Pinecone
- search_web: Live internet search via Tavily
- calculate: Evaluate math expressions
- get_wikipedia_summary: Fetch Wikipedia article summary

Send

Verification:

GET /tools List Tools

Parameters

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8001/tools' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8001/tools
```

Server response

Code

Details

200

Response body

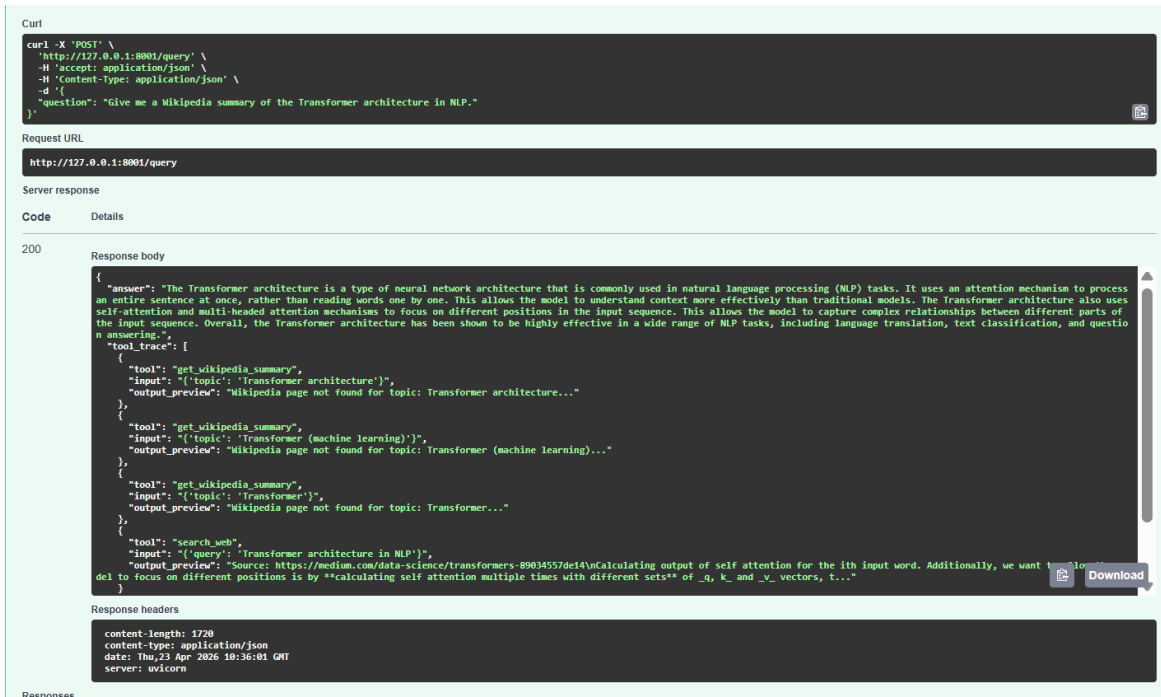
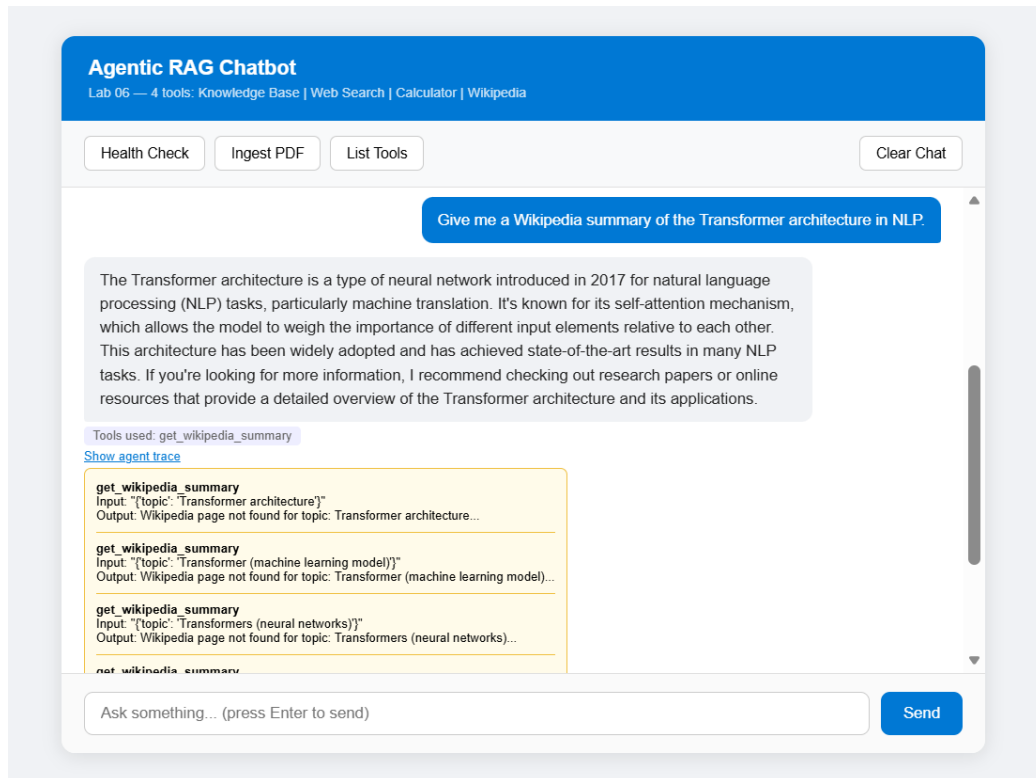
```
{
  "tools": [
    {
      "name": "search_knowledge_base",
      "description": "Search PDF knowledge base in Pinecone"
    },
    {
      "name": "search_web",
      "description": "Live internet search via Tavily"
    },
    {
      "name": "calculate",
      "description": "Evaluate math expressions"
    },
    {
      "name": "get_wikipedia_summary",
      "description": "Fetch Wikipedia article summary"
    }
  ]
}
```

Download

Response headers

Lab task 06

Test 1:



Lab task 06

Test 2:

Agentic RAG Chatbot

Lab 06 — 4 tools: Knowledge Base | Web Search | Calculator | Wikipedia

Health CheckIngest PDFList ToolsClear Chat

What is Python programming language according to Wikipedia?

Python is a high-level, interpreted programming language that is widely used for various purposes such as web development, scientific computing, data analysis, artificial intelligence, and more. It was created by Guido van Rossum and is known for its simplicity, readability, and ease of use. Python is free and open-source, and it runs on multiple platforms including Windows, macOS, and Linux. It has a large and active community, with numerous libraries and frameworks available for different applications. Python is also a popular teaching language, and it's often used in introductory programming courses due to its simplicity and versatility.

Tools used: search_web, get_wikipedia_summary

[Show agent trace](#)

get_wikipedia_summary
Input: "{topic: 'Python programming language'}"
Output: Wikipedia page not found for topic: Python programming language...

get_wikipedia_summary
Input: "{topic: 'Python'}"
Output: Wikipedia page not found for topic: Python...

get_wikipedia_summary
Input: "{topic: 'Python programming'}"
Output: Wikipedia page not found for topic: Python programming...

Send

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8001/query' \
  -H 'accept: application/json' \
  -H 'content-type: application/json' \
  -d '{
    "question": "What is Python programming language according to Wikipedia?"
  }'
```

Request URL

```
http://127.0.0.1:8001/query
```

Server response

Code	Details
200	Response body<pre>{ "answer": "Python is a high-level, interpreted programming language that is widely used for various purposes such as web development, scientific computing, data analysis, artificial intelligence, and more. It was created by Guido van Rossum and is known for its simplicity, readability, and ease of use. Python is free and open-source, and it runs on multiple platforms including Windows, macOS, and Linux. It has a large and active community, and it is widely used in many industries and fields. Python is also a popular teaching language, and it is often used in introductory programming courses due to its simplicity and ease of use.", "tool_trace": [{ "tool": "get_wikipedia_summary", "input": "{topic: 'Python programming language'}", "output_preview": "Wikipedia page not found for topic: Python programming language..." }, { "tool": "get_wikipedia_summary", "input": "{topic: 'Python'}", "output_preview": "Wikipedia page not found for topic: Python..." }, { "tool": "search_web", "input": "{query: 'Python programming language'}", "output_preview": "Source: https://en.wikipedia.org/wiki/Python_(programming_language) [Aragon\u00e9s](https://an.wikipedia.org/wiki/Python_(Python - Aragonese)). * [Catal\u00e0](https://ca.wikipedia.org/wiki/Python_(Python - Catalan)). * [Cebuano](https://ceb.wikipedia.org/wiki/Python_(programming_language)) Python (programm..." }], "tools_used": ["search_web", "get_wikipedia_summary"] }</pre>

Response headers

```
content-length: 1411
content-type: application/json
date: Thu, 23 Apr 2026 10:40:06 GMT
server: uvicorn
```

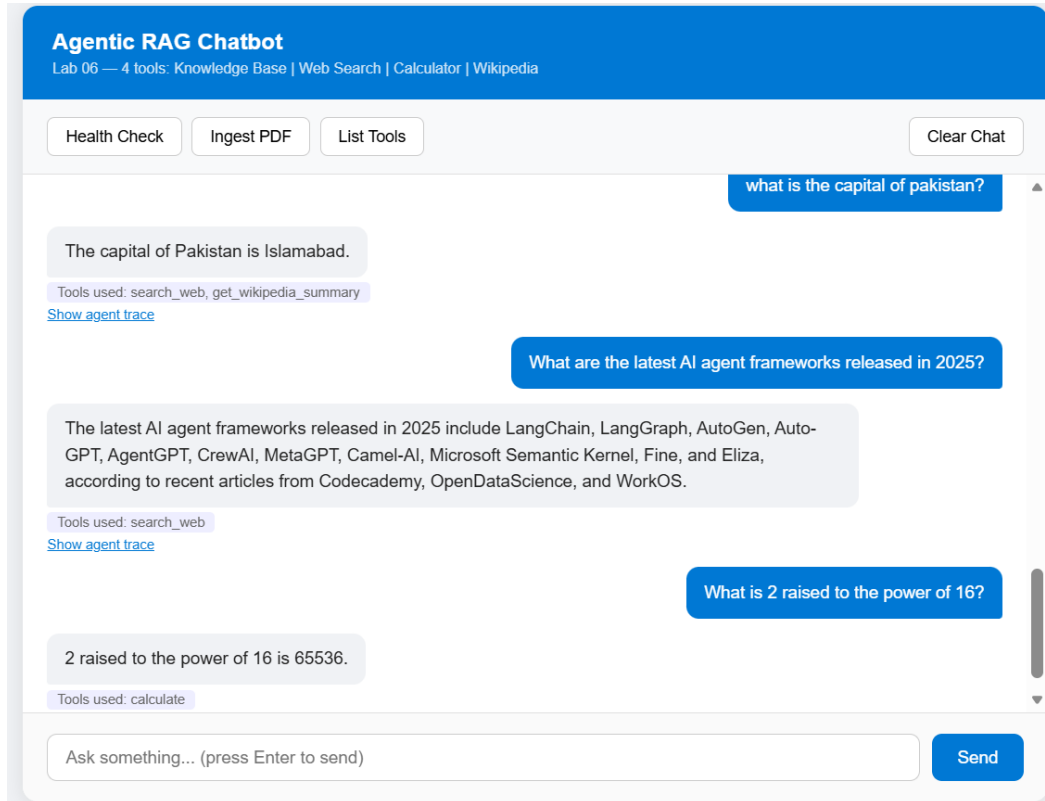
Page 4 | 13

Lab task 06

Task 2:

Add Tool Call Logging

Chat:



Lab task 06

Agentic RAG Chatbot

Lab 06 — 4 tools: Knowledge Base | Web Search | Calculator | Wikipedia

Health CheckIngest PDFList ToolsClear Chat

What is 5 multiplied by 365?

5 multiplied by 365 is 1825.
Tools used: calculate
Show agent trace

What is machine learning according to Wikipedia?

Machine learning is a subfield of artificial intelligence that uses algorithms trained on data sets to create models capable of performing tasks that would otherwise only be possible for humans, such as categorizing images, analyzing data, or predicting price fluctuations. It refers to the general use of algorithms and data to create autonomous or semi-autonomous machines, and is concerned with training models to allow computers to mimic human thought and decision making without explicit programming.
Tools used: search_web, get_wikipedia_summary
Show agent trace

What are the latest AI trends in 2026?

Send

Agentic RAG Chatbot

Lab 06 — 4 tools: Knowledge Base | Web Search | Calculator | Wikipedia

Health CheckIngest PDFList ToolsClear Chat

Tools used: search_web, get_wikipedia_summary
Show agent trace

What are the latest AI trends in 2026?

The latest AI trends in 2026 include the rise of multi-agent systems, agentic AI, domain-specific models, and the application of foundational AI principles in organizations. These trends are expected to shape enterprise transformation and reinvent IT.
Tools used: search_web
Show agent trace

What are the latest Machine Learning trends in 2026?

The latest Machine Learning trends in 2026 include Generative Artificial Intelligence, Internet of Things (IoT) devices, Human-AI collaboration, AI learning shifts, Agent-Based AI, Multimodal AI skills, Agentic AI, MLOps, LLMOps, AgentOps, and explainable AI. These trends are expected to shape the future of Machine Learning and have a significant impact on various industries.
Tools used: search_web
Show agent trace

Send

Lab task 06

Tool_log.txt

	requirements.txt	.env	main.py	tool_log.txt	rag-chatbot.html
tool_log.txt					
1	2026-04-22 21:44:57	search_knowledge_base	query='importance of making something people want'	success	
3	2026-04-22 21:44:57	search_knowledge_base	query='importance of making something people want'	success	
4	2026-04-22 21:45:43	search_knowledge_base	query='importance of making something people want'	success	
5	2026-04-22 21:46:29	search_web	query='latest AI agent frameworks released in 2025'	success	
6	2026-04-22 21:48:16	search_web	query='latest AI agent frameworks 2025'	success	
7	2026-04-22 21:49:18	calculate	query='2 ** 16'	success	
8	2026-04-22 21:51:40	calculate	query='2 ** 16'	success	
9	2026-04-22 21:54:01	calculate	query='2 ** 16'	success	
10	2026-04-22 21:55:16	search_knowledge_base	query='making something people want'	success	
11	2026-04-22 21:56:06	search_knowledge_base	query='making something people want'	success	
12	2026-04-22 21:57:57	search_knowledge_base	query='making something people want'	success	
13	2026-04-22 21:59:43	calculate	query='2 + 2'	success	
14	2026-04-22 22:02:15	search_knowledge_base	query='making something people want'	success	
15	2026-04-22 22:06:59	search_knowledge_base	query='making something people want'	success	
16	2026-04-22 22:13:15	search_knowledge_base	query='making something people want'	success	
17	2026-04-22 22:13:20	search_knowledge_base	query='making something people want'	success	
18	2026-04-22 22:13:26	search_knowledge_base	query='making something people want'	success	
19	2026-04-22 22:13:46	search_knowledge_base	query='making something people want'	success	
20	2026-04-23 14:54:37	calculate	query='2 + 2'	success	
21	2026-04-23 15:32:01	calculate	query='2 + 2'	success	
22	2026-04-23 15:36:09	search_web	query='Transformer architecture in NLP'	success	
23	2026-04-23 15:40:10	search_web	query='Python programming language'	success	
24	2026-04-23 15:45:00	search_web	query='capital of Pakistan'	success	
25	2026-04-23 15:45:32	search_web	query='latest AI agent frameworks released in 2025'	success	
26	2026-04-23 15:46:02	calculate	query='2 ** 16'	success	
27	2026-04-23 15:47:55	search_knowledge_base	query='importance of making something people want'	success	
28	2026-04-23 15:48:54	calculate	query='5 * 365'	success	
29	2026-04-23 15:49:10	search_web	query='machine learning definition'	success	
30	2026-04-23 15:49:17	search_web	query='latest AI trends 2026'	success	
31	2026-04-23 15:50:20	search_web	query='latest Machine Learning trends 2026'	success	

Task 3:

Domain Knowledge Base and Research Reflection

New PDF added:

LAB06_AGENTIC_RAG
> __pycache__
> venv
⚙ .env
📄 domain.pdf
🔗 main.py
📄 paul_graham.pdf
🔗 rag-chatbot.html
≡ requirements.txt
≡ tool_log.txt

Lab task 06

New PDF Ingested:

Agentic RAG Chatbot

Lab 06 — 4 tools: Knowledge Base | Web Search | Calculator | Wikipedia

Health Check

Ingest PDF

List Tools

Clear Chat

Welcome! Checking server connection...

Status: ok | Vectors: 24 | Web Search: true | LLM: llama-3.1-8b-instant

PDF ingested successfully!
Chunks: 19
File: domain.pdf

Ask something... (press Enter to send)

Send

Vector Count

lab06-agentic-rag ⓘ

...

Connect

Record count 51

Region  us-east-1

Model llama-text-embed-v2

Host lab06-agentic-rag-edvu422.svc.aped-4627-b74a...



BROWSER

METRICS

NAMESPACES (2)

CONFIGURATION

Namespaces

Namespaces let you partition an index into groupings for multi-tenancy or query speed. [Learn more.](#)



Filter namespaces

Name	Records	Actions
paul_graham_pdf	24	...
speech_paralinguistics	27	...

Lab task 06

Tests:

Agentic RAG Chatbot

Lab 06 — 4 tools: Knowledge Base | Web Search | Calculator | Wikipedia

Health Check Ingest PDF List Tools Clear Chat

What are the three tasks in SpeechParaling-Bench?

The three tasks in SpeechParaling-Bench are fine-grained control, intra-utterance variation, and context-aware adaptation.

Tools used: search_knowledge_base
[Show agent trace](#)

search_knowledge_base
Input: "[query]: 'SpeechParaling-Bench tasks']"
Output: SpeechParaling-Bench features three progressively challenging tasks: fine-grained control, intra-utterance variation, and context-aware adaptation. Sources: [Page 1, score=0.852] | [Page 9, score=0.851] | [Page 5, score=0.839]...

What percentage of errors are caused by neglecting paralinguage?

According to the available sources, 43.3% of errors are caused by neglecting paralinguage.

Tools used: search_knowledge_base
[Show agent trace](#)

search_knowledge_base
Input: "[query]: 'percentage of errors caused by neglecting paralinguage']"
Output: 43.3% Sources: [Page 9, score=0.76] | [Page 5, score=0.755] | [Page 7, score=0.736]...

Ask something... (press Enter to send) Send

Agentic RAG Chatbot

Lab 06 — 4 tools: Knowledge Base | Web Search | Calculator | Wikipedia

Health Check Ingest PDF List Tools Clear Chat

What percentage of errors are caused by neglecting paralinguage?

According to the available sources, 43.3% of errors are caused by neglecting paralinguage.

Tools used: search_knowledge_base
[Show agent trace](#)

search_knowledge_base
Input: "[query]: 'percentage of errors caused by neglecting paralinguage']"
Output: 43.3% Sources: [Page 9, score=0.76] | [Page 5, score=0.755] | [Page 7, score=0.736]...

How does the pairwise evaluation pipeline work in SpeechParaling-Bench?

The pairwise evaluation pipeline in SpeechParaling-Bench works by comparing candidate responses against a fixed baseline, reformulating evaluation as relative preference rather than absolute scoring, which mitigates the inherent subjectivity of paralinguistic assessment, resulting in more stable and scalable evaluation without costly human annotation.

Tools used: search_knowledge_base
[Show agent trace](#)

Ask something... (press Enter to send) Send

Lab task 06

Agentic RAG Chatbot

Lab 06 — 4 tools: Knowledge Base | Web Search | Calculator | Wikipedia

Health Check

Ingest PDF

List Tools

Clear Chat

Tools used: search_knowledge_base

[Show agent trace](#)

search_knowledge_base
Input: "[query: 'percentage of errors caused by neglecting paralinguistics']"
Output: 43.3% Sources: [Page 9, score=0.76] | [Page 5, score=0.755] | [Page 7, score=0.736]...

How does the pairwise evaluation pipeline work in SpeechParaling-Bench?

The pairwise evaluation pipeline in SpeechParaling-Bench works by comparing candidate responses against a fixed baseline, reformulating evaluation as relative preference rather than absolute scoring, which mitigates the inherent subjectivity of paralinguistic assessment, resulting in more stable and scalable evaluation without costly human annotation.

Tools used: search_knowledge_base

[Show agent trace](#)

search_knowledge_base
Input: "[query: 'pairwise evaluation pipeline SpeechParaling-Bench']"
Output: We propose an automated pairwise evaluation framework that compares candidate responses against a fixed baseline. By reformulating evaluation as relative preference rather than absolute scoring, this approach mitigates the inherent subjectivity of paralinguistic assessment, resulting in more stable ...

Ask something... (press Enter to send)

Send

Agentic RAG Chatbot

Lab 06 — 4 tools: Knowledge Base | Web Search | Calculator | Wikipedia

Health Check

Ingest PDF

List Tools

Clear Chat

Output: We propose an automated pairwise evaluation framework that compares candidate responses against a fixed baseline. By reformulating evaluation as relative preference rather than absolute scoring, this approach mitigates the inherent subjectivity of paralinguistic assessment, resulting in more stable ...

What is paralinguistics according to Wikipedia?

Paralinguistics refers to the ability to convey meaning and emotion through non-verbal cues, such as tone of voice, facial expressions, and body language.

Tools used: get_wikipedia_summary, search_knowledge_base

[Show agent trace](#)

get_wikipedia_summary
Input: "{topic: 'paralinguistics'}"
Output: Wikipedia page not found for topic: paralinguistics...

search_knowledge_base
Input: "[query: 'paralinguistics']"
Output: The ability to convey meaning and emotion through non-verbal cues, such as tone of voice, facial expressions, and body language. Sources: [Page 11, score=0.792] | [Page 5, score=0.79] | [Page 12, score=0.781]...

What are the latest large audio language models in 2026?

Ask something... (press Enter to send)

Send

Lab task 06

Agentic RAG Chatbot

Lab 06 — 4 tools: Knowledge Base | Web Search | Calculator | Wikipedia

Health Check

Ingest PDF

List Tools

Clear Chat

Input: `{ "query": "paralinguistics" }`

Output: Wikipedia page not found for topic: paralinguistics...

search_knowledge_base
Input: `{ "query": "paralinguistics" }`
Output: The ability to convey meaning and emotion through non-verbal cues, such as tone of voice, facial expressions, and body language. Sources: [Page 11, score=0.792] | [Page 5, score=0.79] | [Page 12, score=0.781]...

What are the latest large audio language models in 2026?

According to the latest sources, some of the latest large audio language models in 2026 include MOSS-TTS, MOSS-TTSD, GPT-5.2, Gemini 3.1 Pro, DeepSeek-V3.2, Claude 4.6 Opus, Grok-4, and Mistral Large 2.

Tools used: search_web

[Show agent trace](#)

search_web
Input: `{ "query": "latest large audio language models 2026" }`
Output: Source: https://www.reddit.com/r/LocalLLaMA/comments/1r7bsfd/best_audio_models_feb_2026/
MOSS-TTS: The flagship production model featuring high fidelity and optimal zero-shot voice cloning.
MOSS-TTSD: A spoken dialogue generation Source: <https://explodingtopics.com/blog/list-of-llms-top-50-large->

Ask something... (press Enter to send)

Send

```
requirements.txt  .env  main.py  domain.pdf  paul_graham.pdf  tool_log.txt  rag-chatbot.html

tool_log.txt
42 2026-04-23 16:50:27 | search_knowledge_base | query='SpeechParaling-Bench tasks' | success
43 2026-04-23 16:51:00 | search_knowledge_base | query='percentage of errors in situational dialogue caused by neglectin
44 2026-04-23 16:51:46 | search_knowledge_base | query='SpeechParaling-Bench pairwise comparison evaluation' | success
45 2026-04-23 16:52:27 | get_wikipedia_summary | query='paralinguistics' | failed
46 2026-04-23 16:52:29 | search_web | query='paralinguistics definition' | success
47 2026-04-23 16:52:58 | search_web | query='latest large audio language models 2026' | success
48 2026-04-23 16:58:47 | search_knowledge_base | query='SpeechParaling-Bench tasks' | success
49 2026-04-23 16:59:31 | search_knowledge_base | query='percentage of errors caused by neglecting paralanguage' | succes
50 2026-04-23 17:00:12 | search_knowledge_base | query='pairwise evaluation pipeline SpeechParaling-Bench' | success
51 2026-04-23 17:02:25 | search_knowledge_base | query='paralinguistics' | success
52 2026-04-23 17:02:33 | search_web | query='latest large audio language models 2026' | success
53 2026-04-23 17:16:12 | search_knowledge_base | query='paralinguistics' | success
54
```

Research reflection

(a) What questions worked well with the KB tool and why?

The questions that worked best were the ones that asked about specific things directly from the paper. For example:

- *"What are the three tasks in SpeechParaling-Bench?"* This worked great because the paper clearly lists Paralanguage Control, Dynamic Variation, and Situational Adaptation. The KB found it instantly.
- *"What percentage of errors are caused by neglecting paralanguage?"* This worked well because 43.3% is a very specific number that only exists in this paper, so the KB had no trouble finding the right chunk.
- *"How does the pairwise evaluation pipeline work?"* This worked because the paper has a dedicated section explaining it, so the retrieved chunks were highly relevant.

These questions worked well because they asked for **specific facts, numbers, or methods** that are unique to the paper. When a question uses terms that appear directly in the document (like "SpeechParaling-Bench" or "pairwise evaluation"), the embedding model can match them easily.

(b) Which questions caused the agent to incorrectly choose the wrong tool, and how would you fix the docstring?

One question that went wrong was: *"What is paralinguistics?"*

Instead of searching the KB, the agent used **Wikipedia** and **web search**. This happened because "paralinguistics" sounds like a general concept that Wikipedia would know about and the KB tool's description didn't specifically mention that this topic is covered in the uploaded paper.

How i fixed it: I updated the KB tool's docstring to list specific domain terms so the agent knows to look there first. For example, add: *"Use this tool for questions about paralinguistics, LALMs, SpeechParaling-Bench, pairwise evaluation, Paralanguage Control, Dynamic Variation, or Situational Adaptation, even if the concept also exists as general knowledge."*

By listing these specific terms in the docstring, the agent matched user questions to the KB tool instead of defaulting to Wikipedia.

(c) How would you deploy this agent in a production setting for your research domain?

If I were deploying this agent for real in a speech AI research setting,

Lab task 06

Hosting: I would move the FastAPI backend from my laptop to a cloud server like AWS or Google Cloud, packaged in Docker so it runs consistently anywhere.

Security: API keys are in a `.env` file right now, which is fine for a lab but not for production. I would store them in a proper secrets manager like AWS Secrets Manager so they are never exposed.

Scaling: The current setup handles one user at a time. For multiple researchers using it simultaneously, I would put a load balancer in front of multiple server instances.

Knowledge Base Updates: Speech AI is a fast-moving field, new papers come out every week. I'd set up an automated pipeline that regularly ingests new research papers into Pinecone so the KB stays current.

Monitoring: I would extend the current `tool_log.txt` to a proper dashboard (like Datadog) so I can see which tools are being used, which questions fail, and how fast responses are.

Better Tool Routing: After running it for a while, I would analyze the logs to find cases where the agent picked the wrong tool and keep improving the docstrings to make routing more accurate over time.

- END -